

SSN College of Engineering, Kalavakkam
Department of Computer Science and Engineering
V Semester - CSE 'A'
UCS2511 NETWORKS LAB
Exercise 3
File Transfer Using TCP

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Aim:

File Transfer In C

server.c

```
#include <stdio.h>
#include <sys/socket.h>
#include <netinet/in.h>
#include <stdlib.h>
#include <unistd.h>
#include <dirent.h>
#include <string.h>
#include <libgen.h>
#include <sys/stat.h>
#include <fcntl.h>
```

```
char filepath[PATH_MAX] = " ";
```

```
void FileTransfer(const char *path, const char *filename) {
    DIR *currentdirectory;
    struct stat s;
    struct dirent *file;

    if ((currentdirectory = opendir(path)) == NULL) {
        perror("opendir");
        return;
    }
```

```
};
```

```
while ((file = readdir(currentdirectory)) != NULL) {  
    char fullpath[PATH_MAX]; // Use a large enough buffer  
    strcpy(fullpath, path);
```

```
    strcat(fullpath, "/");  
    strcat(fullpath, file->d_name);
```

```
    stat(fullpath, &s);  
    if (!strcmp(file->d_name, filename)) {
```

```
        printf("Found: %s\n", fullpath);  
        strcpy(filepath, fullpath);  
        break;
```

```
}
```

```
if (S_ISDIR(s.st_mode)) {  
    if (strcmp(file->d_name, ".") != 0 && strcmp(file-
```

```
>d_name, ".") != 0) {  
        //printf("\n%s", fullpath);
```

```
call  
}
```

```
}}
```

```
FileTransfer(fullpath, filename); // Recursive
```

```
    closedir(currentdirectory);  
}
```

```
char* FTP(char* filepath)  
{
```

```
    static char buffer[1024];  
    if(!strcmp(filepath, ""))
```

```
    {  
        strcpy(buffer, "FileNotFound");
```

```
} else
```

```
{  
    int fd = open(filepath, O_RDONLY);  
    struct stat s;  
    stat(filepath, &s);
```

```
    read(fd, buffer, s.st_size);
```

```
}
```

```
    return buffer;

}

int main(int argc, char** argv)
{

    int socket_id, client_id;
    struct sockaddr_in server,client;
    char filename[20];
    char* data;

    if(argc<2) {

        printf("Usage: %s <port>\n",argv[0]);
        exit(1);

    }

    socket_id = socket(AF_INET,SOCK_STREAM,0);

    server.sin_family = AF_INET;
    server.sin_port = htons(atoi(argv[1]));
    server.sin_addr.s_addr = INADDR_ANY;


    bind(socket_id, (struct sockaddr *) &server, sizeof(server));
    printf("Binding...\n");

    listen(socket_id, 3);
    printf("listening...\n");
```

```
socklen_t size = sizeof(client);
```

```
client_id = accept(socket_id,(struct sockaddr*) &client,  
&size);
```

```
if(client_id < 0)  
{
```

```
    perror("Connection Failed\n");  
}
```

```
printf("Connected...\n");
```

```
int bytes_read = recv(client_id,filename,sizeof(filename),0);  
filename[bytes_read] = '\0';
```

```
FileTransfer("/home/",filename);  
data = FTP(filepath);
```

```
send(client_id,data,strlen(data),0);
```

```
close(client_id);  
close(socket_id);
```

```
return 0; } client.c
```

```
#include <stdio.h>  
#include <sys/socket.h>  
#include <netinet/in.h>  
#include <stdlib.h>  
#include <unistd.h>  
#include <string.h>
```

```
int main(int argc, char** argv)
{

    int socket_id;
    struct sockaddr_in client;

    char data[1024];

    if(argc<3) {

        printf("Usage: %s <port> <filename>\n",argv[0]);
        exit(1);

    }
    socket_id = socket(AF_INET,SOCK_STREAM,0);

    client.sin_family = AF_INET;
    client.sin_port = htons(atoi(argv[1]));
    client.sin_addr.s_addr = INADDR_ANY;

    connect(socket_id, (struct sockaddr*) &client,
sizeof(client));

    printf("Connected to server!\n");
    send(socket_id, argv[2], strlen(argv[2]),0);
    recv(socket_id, data, sizeof(data),0);
    printf("Data: %s",data);
    close(socket_id);

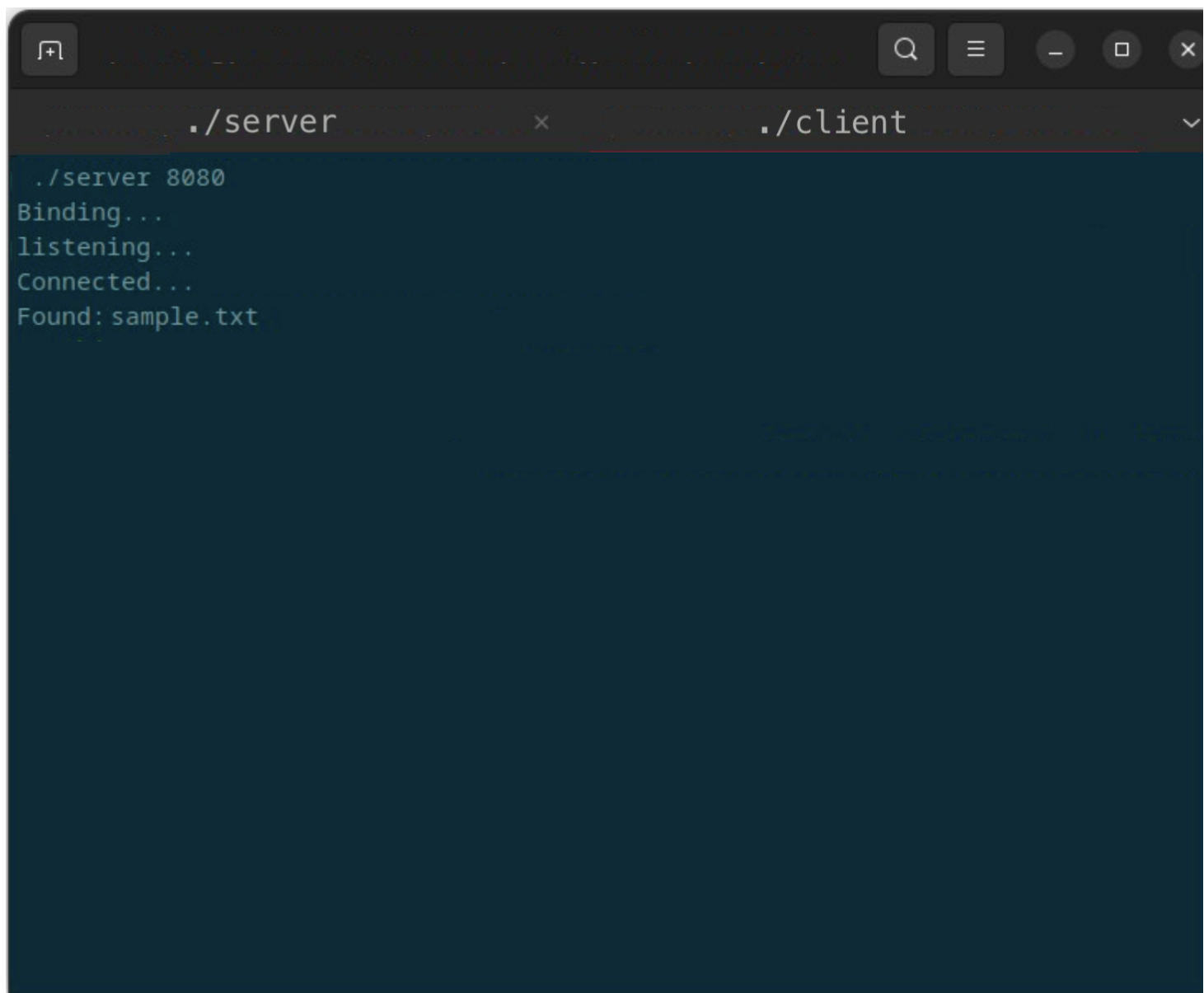
    return 0;
```

```
}
```

sample.txt

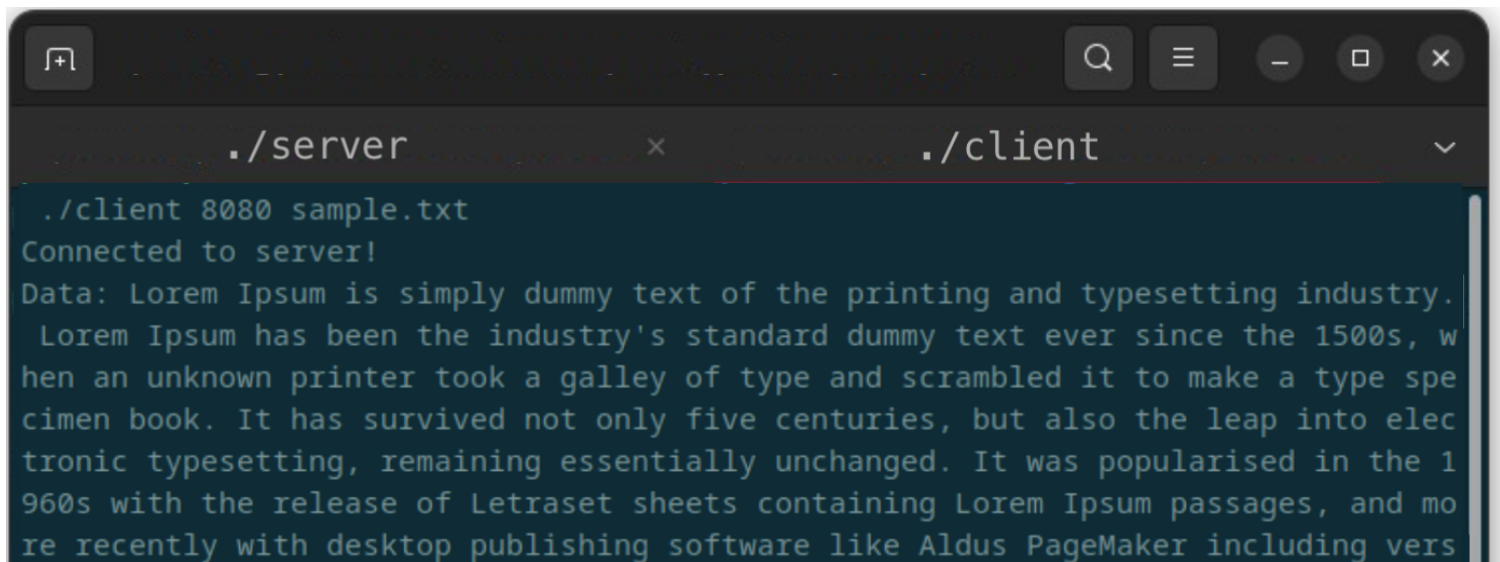
Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book. It has survived not only five centuries, but also the leap into electronic typesetting, remaining essentially unchanged. It was popularised in the 1960s with the release of Letraset sheets containing Lorem Ipsum passages, and more recently with desktop publishing software like Aldus PageMaker including versions of Lorem Ipsum.

Output:

A terminal window with a dark background and light-colored text. The window has a title bar at the top with standard macOS window controls (red, yellow, green buttons) and a search bar. Below the title bar, there are two tabs: './server' and './client'. The './server' tab is active, showing the following text: './server 8080', 'Binding...', 'listening...', 'Connected...', and 'Found: sample.txt'. The './client' tab is inactive. The text is in a monospaced font.

```
./server 8080
Binding...
listening...
Connected...
Found: sample.txt
```

O/P:



A terminal window with a dark background and light-colored text. The window has a title bar with standard macOS window controls (red, yellow, green buttons) and a search bar. Below the title bar, there are two tabs: `./server` and `./client`. The `./client` tab is active. The terminal output shows the following sequence of commands and responses:

```
./client 8080 sample.txt
Connected to server!
Data: Lorem Ipsum is simply dummy text of the printing and typesetting industry.
Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, w
hen an unknown printer took a galley of type and scrambled it to make a type spe
cimen book. It has survived not only five centuries, but also the leap into elec
tronic typesetting, remaining essentially unchanged. It was popularised in the 1
960s with the release of Letraset sheets containing Lorem Ipsum passages, and mo
re recently with desktop publishing software like Aldus PageMaker including vers
```

Learning Outcomes:

I learned how to transfer files using socket programming.